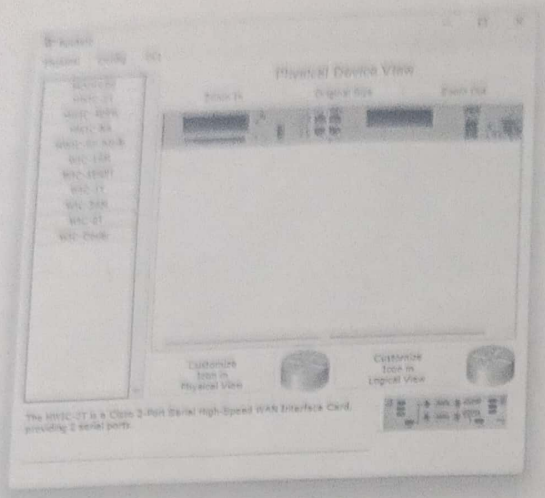
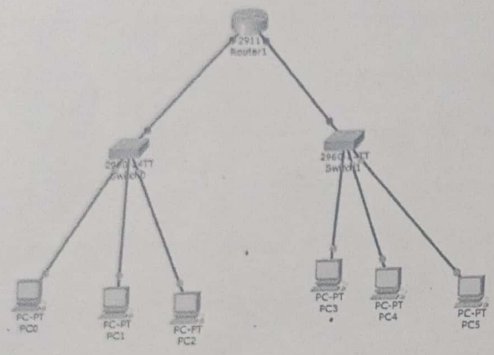





```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```

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EXP:- Implementation of Subnet in Cisco Packet

(a) Design Multiple Subnets with a suitable No of Hosts

Aim:-
To implement Subnetting by divide a large networks into multiple smaller subnetworks according to the required no of hosts and to study the allocation of IP addresses in Cisco Packet Tracer.

Steps:-
1. Open cisco Packet Tracer and create a new project.
2. Design the network topology by placing the required network devices such as routers, Switches and Pcs.
3. Select a suitable network address for the experiment
4. Calculate the no of Subnets required.
5. Determine the no of hosts needed in each Subnet
6. Calculate the Subnet mask using Subnetting techniques.
7. Divide the network into multiple Subnets.
8. Assign a Seperate network address to each Subnet.
9. Verify the Subnet calculations and host ranges
10. Save the network topology.

Result:- Thus, multiple Subnets were Successfully created according to the required no of hosts and the network address was efficiently divided into smaller subnetworks.

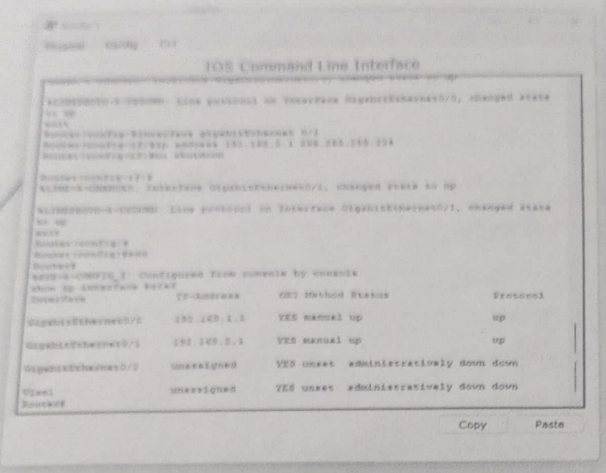
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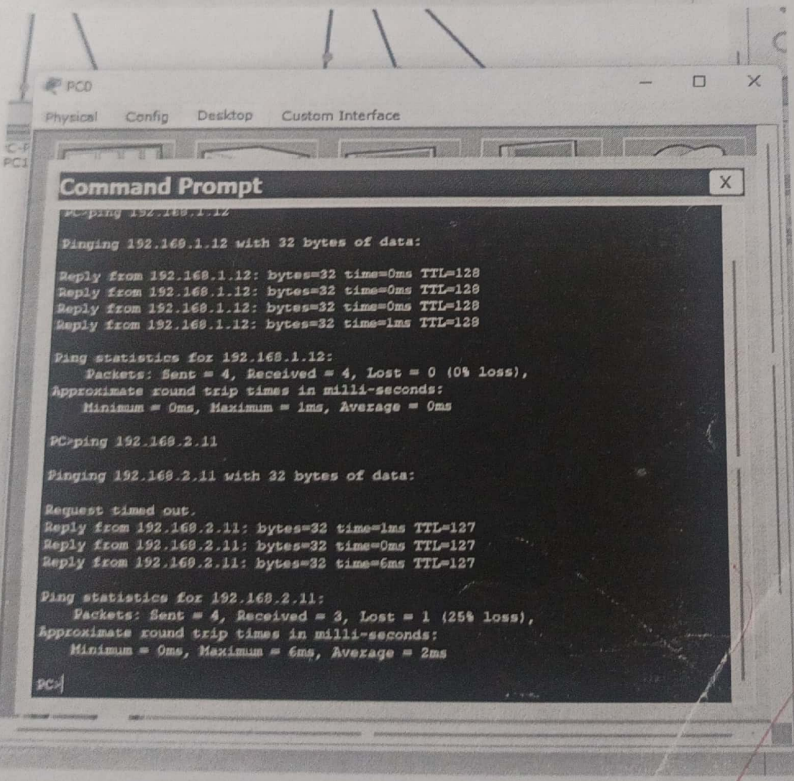
Ch2 Assign Ip
All Subnets

Aim:- To a
To all d
Subnets and
b/w the d

Steps:-
1. Place a
topology
2. Connect H
using appro
3. Connect
4. Assign d
mask to
5. Configure
6. Assign
Computers
7. Configure
device.
8. Assign
to Rout
9. Verify
devices.
10. Save
11. Check
Configura
12. Verify
Properly

Result:-
Successfu
Commu
Netw





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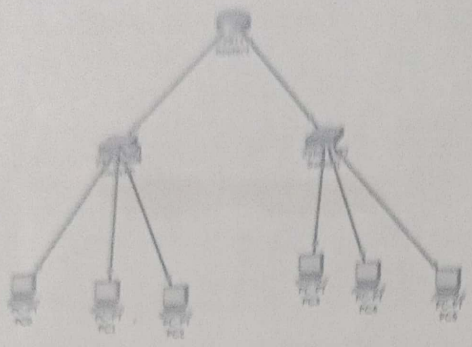
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6) Assign Static Ip Addresses Across All Subnets and Connect the Subnets

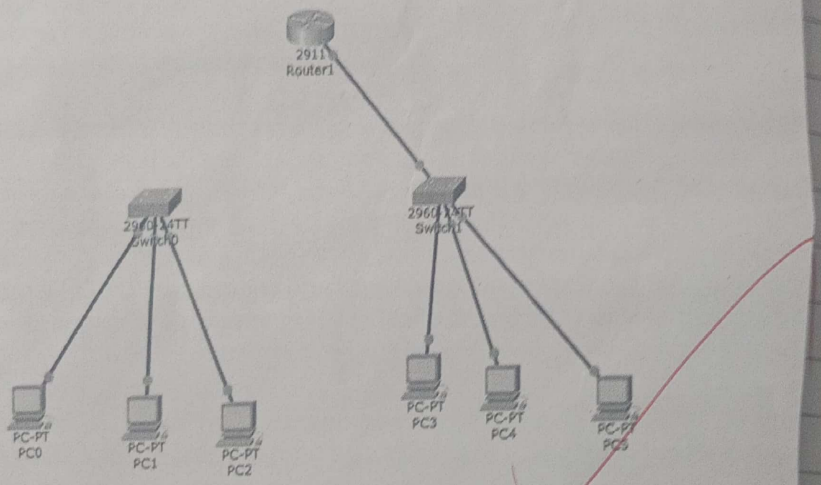
Aim:- To assign Static Ip address to all devices in a different Subnets and establish communication b/w the Subnets through a router.

Steps:-
1. Place a router in the network topology
2. Connect the router to the Switches using appropriate cables.
3. Connect the configure interfaces.
4. Assign an Ip addresses and Subnet mask to each router Interface.
5. Configure the Switches.
6. Assign Static Ip addresses to all Computers in every Subnet.
7. Configure the Subnet mask for each device.
8. Assign the appropriate default gateway to each Computer
9. Verify the Ip Configuration of all devices.
10. Save the router configuration.
11. Check the network topology for configuration errors.
12. Verify that all devices are Properly Connected.

Result:- Thus, Static Ip addresses were Successfully assigned to all devices, and Communication b/w different Subnet through router.



Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC0	PC3	ICMP		0.000	N	6	(edit)	(delete)



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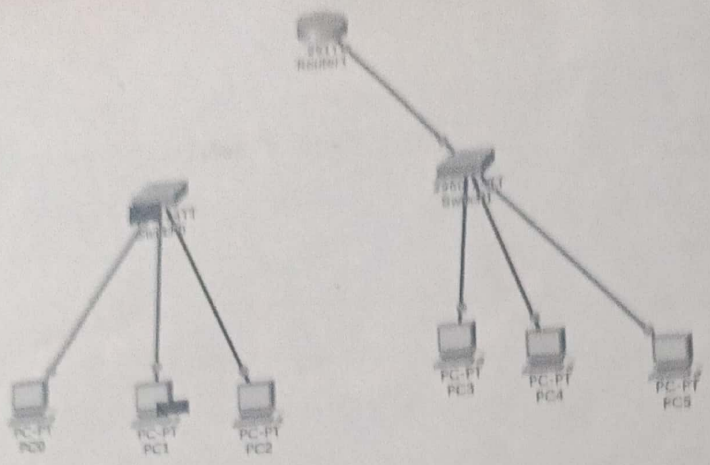
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(c) Simulate Packet Transmission Across the Subnets and Observe the Results (or) When Subnets Connected through Router

Aim:-
To analyze Packet transmission b/w different Subnets when a router is used for communication.

Steps:-
1. Open Simulation Mode in Cisco Packet Tracer
2. Select a Source device from one Subnet.
3. Select a Destination device from another Subnet
4. Generate a Packet using the Simple PDU tool.
5. Observe the movement of Packet through network.
6. Analyze how the router receives the Packet
7. Observe the Forwarding Process Performed by the header.
8. Verify the Packet transmission b/w different Subnets
9. Use the Ping Command to test connectivity
10. Observe the Sucess delivery of Packets
11. Record the Simulation results.
12. Analyze the routing Process.

Result:- Thus Packet were Successfully transmitted b/w Different Subnets through the router. The router received the Packets from the Source Subnet, Processed them and forwarded them to Destination Subnets. Successful Ping responses confirmed Proper Communication b/w all network devices.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)

PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

DHCP Static

IP Address 192.168.1.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server

IPv6 Configuration

DHCP Auto Config Static

IPv6 Address

Link Local Address FE80::2D0:BAFF:FE9C:D809

IPv6 Gateway

IPv6 DNS Server

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(Ao 11) When Subnets are NOT Connected through a router

Aim :- To analyze Packet transmission b/w different Subnets when no router is available.

Steps :-

1. Remove the router from the network topology.
2. Open Simulation Mode
3. Select devices belonging to different Subnets
4. Generate Packets using the Simple PDU tool
5. Observe the Packet transmission.
6. Perform a Ping test b/w devices located in different Subnets
7. Analyze the Packet flow.
8. Observe the Simulation events
9. Identify the Point at which Packet transmission fails
10. Compare the result with the network that uses a router
11. Record the Observations.
12. Analyze the importance of a router in Inter-network Communication.

Result :- Thus Communication b/w devices located in different Subnets failed when a router was not used. The Packets could not reach the destination because there was no routing device to forward them. This experiment demonstrated the importance of routers in enabling Communication b/w different Subnetworks.